Total No. of Questions : 10]

SEAT No. :

P1669

[Total No. of Pages : 3

[5460]-196

T.E.(Information Technology)

DESIGN AND ANALYSIS OF ALGORITHMS

(2012 Pattern) (Semester-II)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5. or Q.6, Q.7 or Q.8 and Q.9 or Q.10.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***

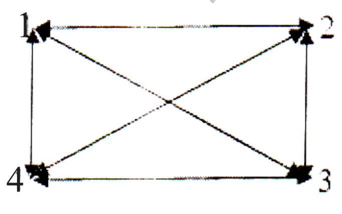
Q1) a) Give the significance of analysis of algorithms. Also compare a priori analysis and a posteriori analysis of algorithms. **[6]**

b) Analyze the time complexity of Strassen's matrix multiplication using divide and conquer strategy. **[4]**

OR

Q2) a) Prove by contradiction that "there exists two irrational numbers x and y such that x^y is rational." **[4]**

b) Find the solution of the following travelling sales person problem using Dynamic approach. **[6]**



$$\begin{pmatrix} 0 & 10 & 15 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{pmatrix}$$

Q3) a) Write and explain greedy method control abstraction for Subset paradigm. **[4]**

b) What is dynamic programming? Is this the optimization technique? Give reasons. What are its drawbacks? Explain memory functions. **[6]**

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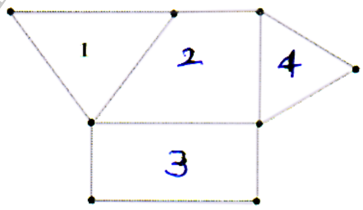
OR

Q4) a) Write and explain Warshall's algorithm to fine transitive closure of a graph. Give time complexity for the same. **[6]**

b) Compare Divide and Conquer strategy and Greedy strategy. **[4]**

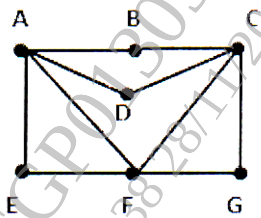
Q5) a) What is a graph coloring problem? Define a chromatic number of a graph. Find the chromatic number of a following map and draw necessary state space tree. **[8]**



b) Solve the 4-Queen's problem. Draw the state space tree for the same. **[8]**

OR

Q6) a) Find Hamiltonian cycles starting from A in given graph. Draw state space tree for the same. **[8]**



b) State and explain the principle of backtracking. Explain the constraints used in backtracking with an example. **[8]**

Q7) a) Solve the following instance of 0/1 knapsack problem by LC branch and bound approach: n=4; M=18, P(l:4)={1 1,11,12,18}, W(l:4)={3,4,6,9}. **[10]**

b) What is LC search? How does it help in finding a solution for branch and bound algorithm? **[8]**

OR

Q8) Write short notes on: **[18]**

a) TSP by Branch and bound strategy

b) Bounding Function

c) Various searching techniques in branch and bound

Q9) a) Give detail proof for clique problem is NP-complete. **[8]**

b) What is the significance of parallel computing? Explain different fixed connection machines for parallel computing. **[8]**

OR

Q10) a) Write non-deterministic algorithm for sorting of elements of an array. What is its complexity? **[8]**

b) Explain the following terms in brief: **[8]**

i) PRAM

ii) EREW PRAM

iii) CRCW PRAM

iv) Speedup of the parallel algorithm

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